

GUI DEVELOPMENT

User experience

WHAT IS A UI?

- Wikipedia:

- *a user interface (UI) is the space where **interactions between humans and machines** occur. The **goal** of this interaction is to **allow effective operation and control** of the machine from the human end, whilst the machine simultaneously **feeds back information** that aids the operators' decision-making process.*
- *a **graphical user interface (GUI)** is a form of user interface that allows users to **interact** with electronic devices through **graphical icons**. ... The actions in a GUI are usually performed through **direct manipulation** of the **graphical elements***

IMPORTANCE OF A GOOD UI

- sports administration



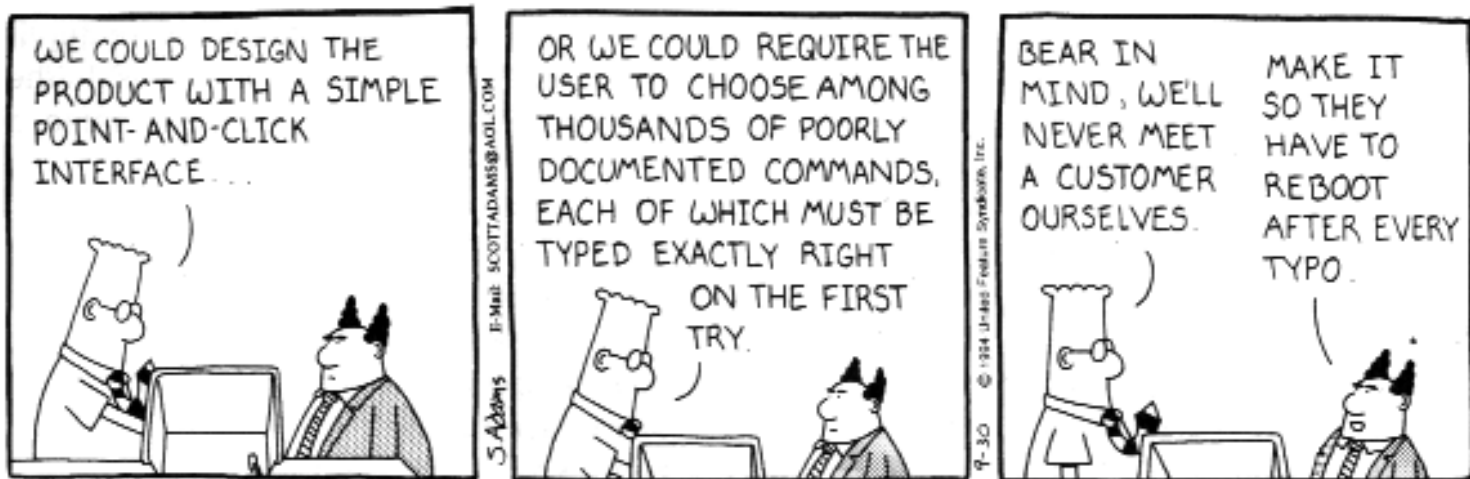
HISTORY

- 50 years ago:
 - Big
 - Expensive
 - Operated by specialists
 - Command line interface (CLI)
- Today:
 - Small
 - Cheap
 - Non-computer specialists too



SYSTEM-CENTERED DESIGN

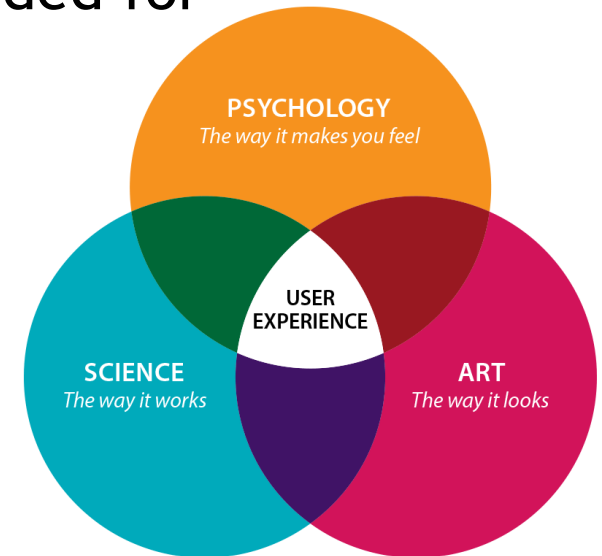
- Functionality comes first
- Good software is more important than ease of use
- User must adapt to the system
- UI is added at the end



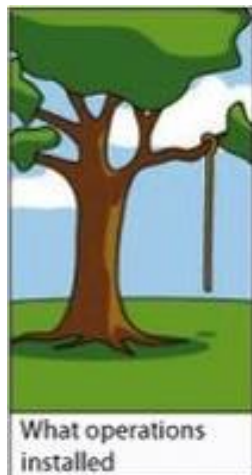
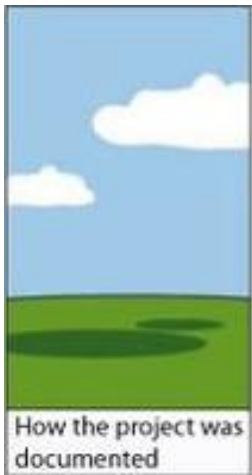
DILBERT reprinted by permission of United Feature Syndicate, Inc.

USER-CENTERED DESIGN

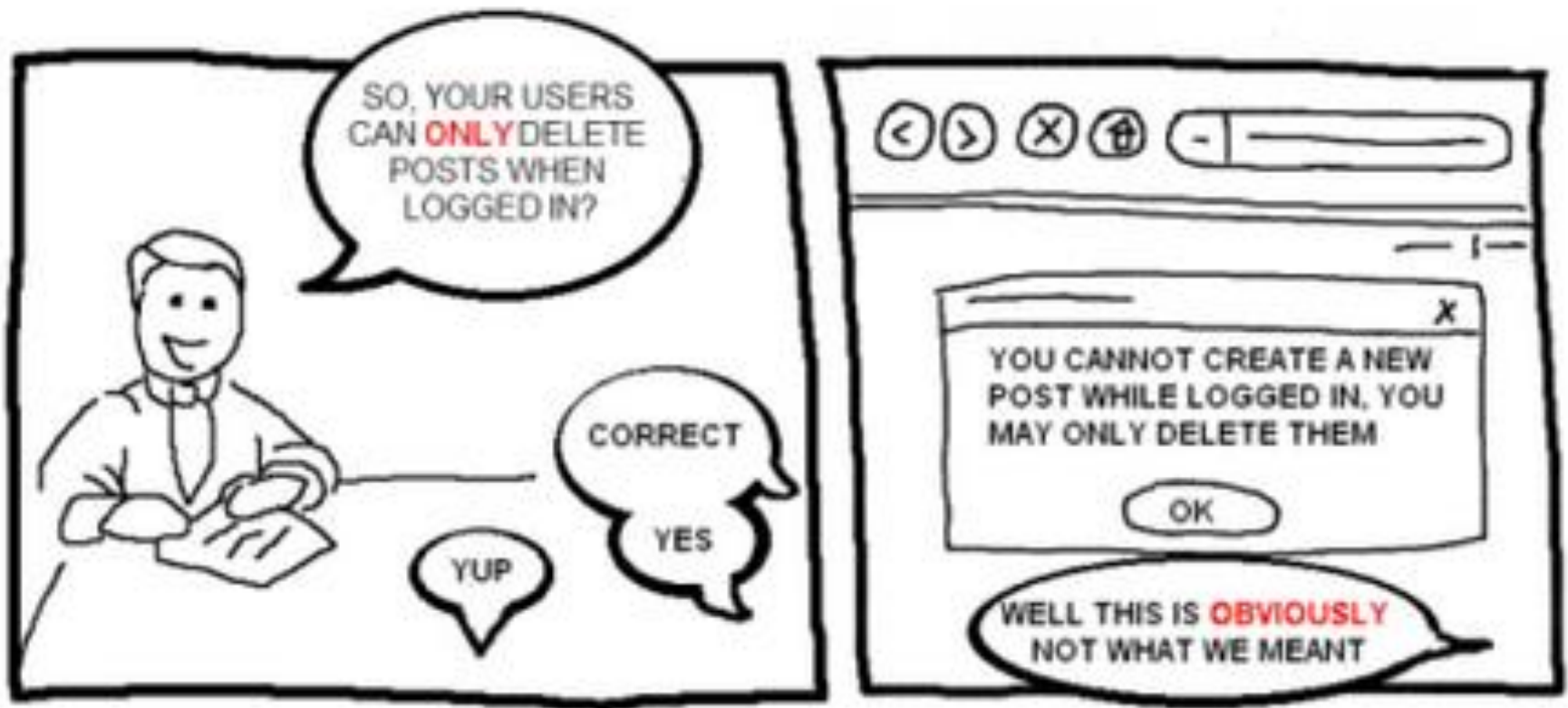
- Involve users in the design process
- good understanding of requirements needed for
 - Tasks
 - Users
- iterative process
- multi-disciplinary
 - computer science (software)
 - Engineering (devices)
 - Ergonomics
 - Graphical design
 - Cognitive psychology (memory, mental models)
 - Sociology (interaction between people)
 - Anthropology (studying people in their working environment)
 - ...



UNDERSTANDING THE REQUIREMENTS



UNDERSTANDING THE REQUIREMENTS



Why are things always so obvious after disaster happens, I think they call that hindsight

UNDERSTANDING THE REQUIREMENTS

- interviews and observations
- workshops and focus groups with end users
- web surveys
- review / test existing solutions
- ...



THE BIG PICTURE

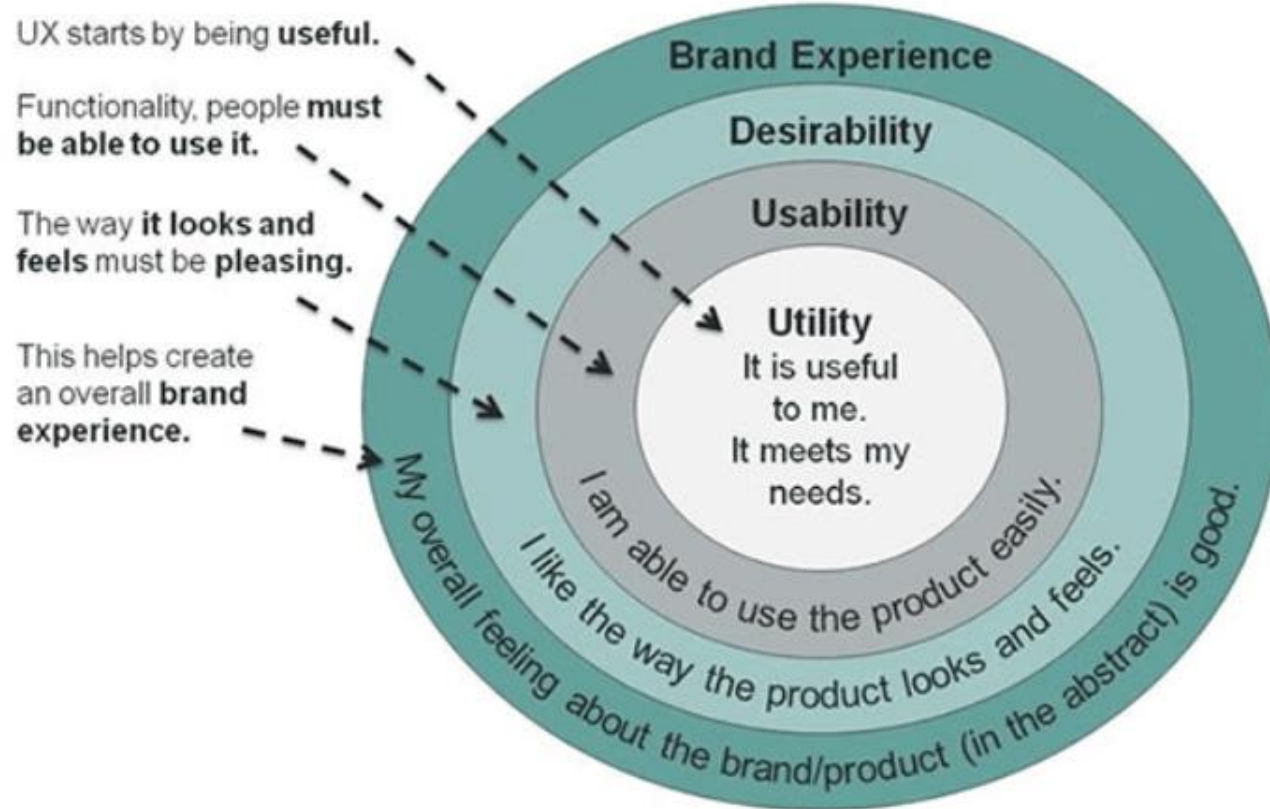


People don't buy products, they buy better versions of themselves.

=> Find out which features really need to be added

ALWAYS consider "the big picture" => what does the customer want to achieve and what is the best way to do it?

USER EXPERIENCE OR UX



Source: User Experience 2008, nnGroup Conference Amsterdam

USER CENTERED DESIGN FOR GOOD UX

Discover

- Observing customers in their workplace
- Listening to problems and suggestions
- Learning what is meaningful



Design



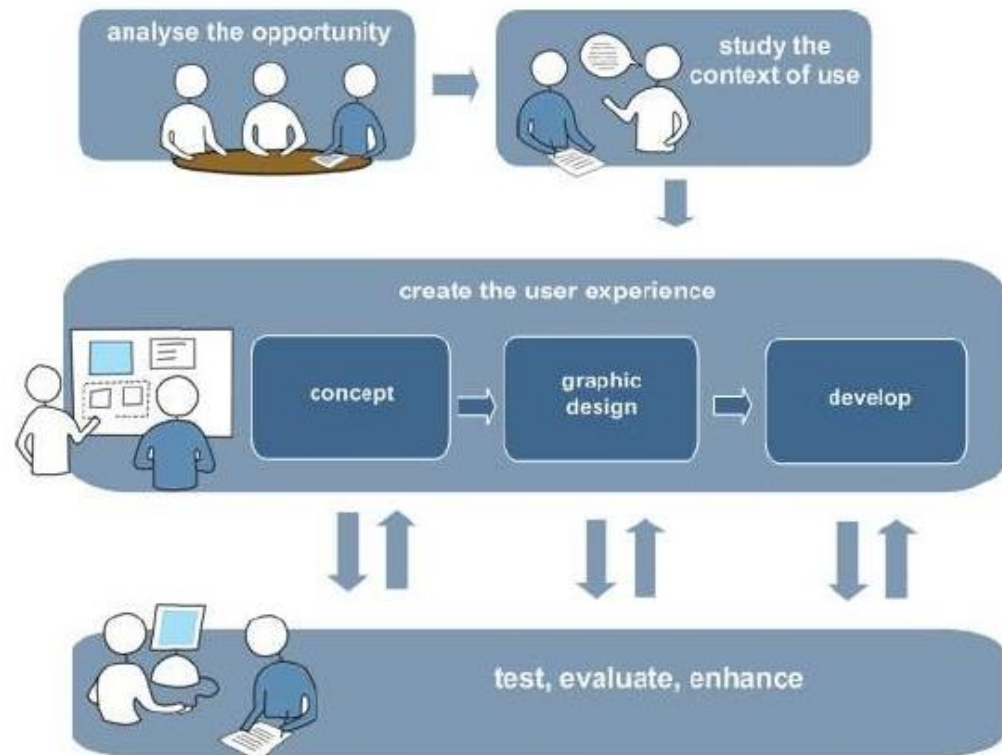
Our Customer

Evaluate

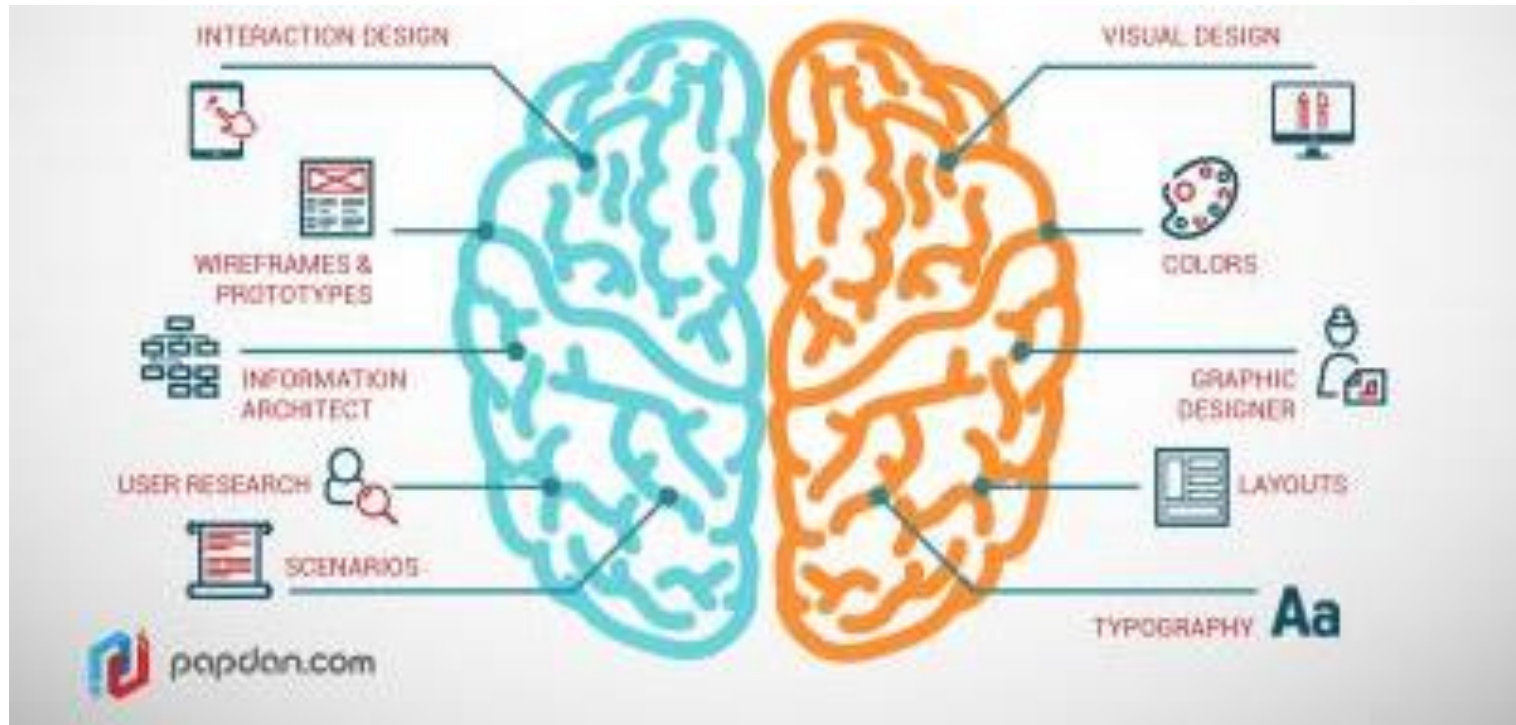
- Reflecting on customer needs
- Brainstorming new ideas
- Sketching design concepts



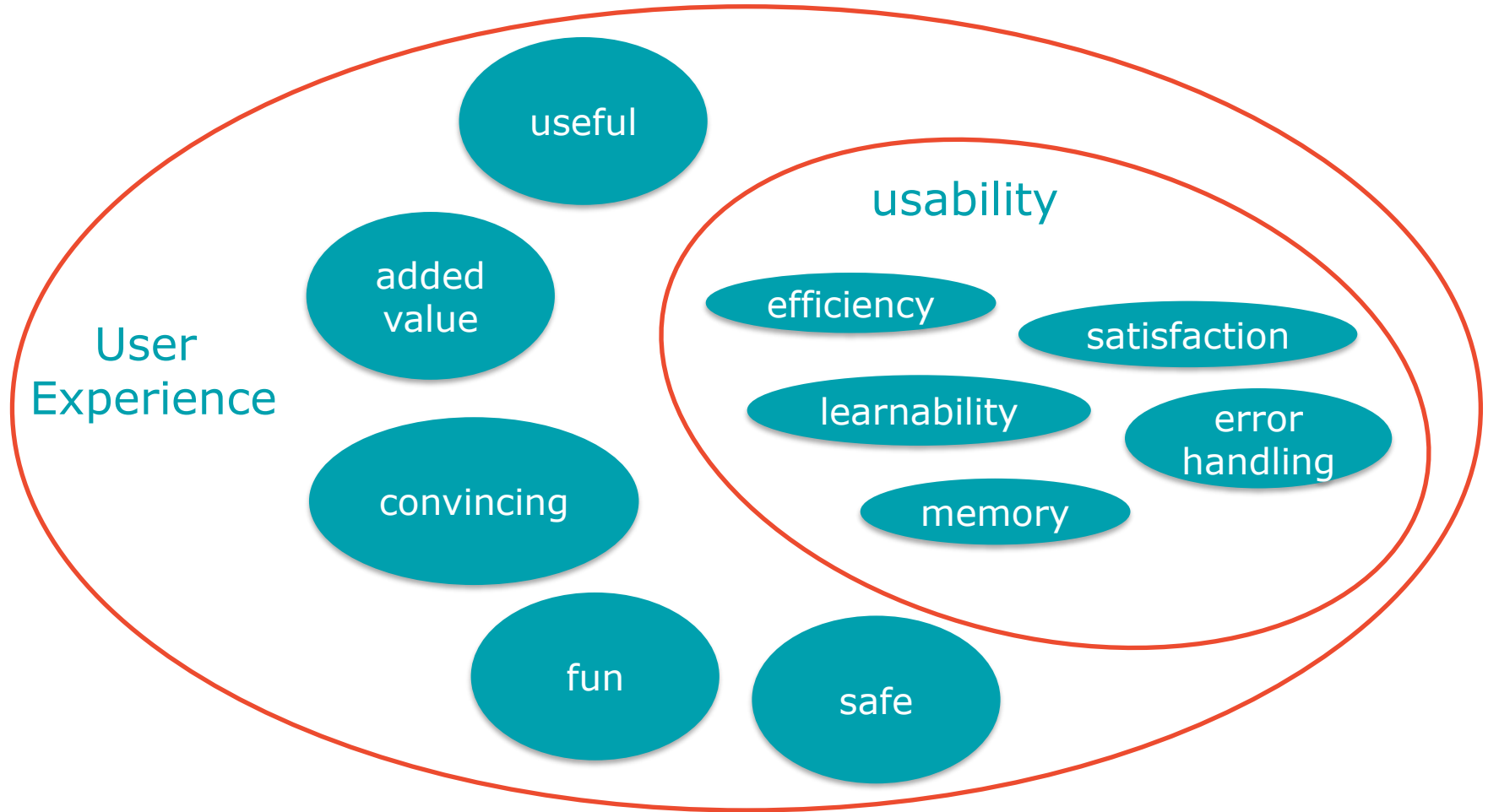
- Testing concepts with customers
- Measuring the results
- Improving designs



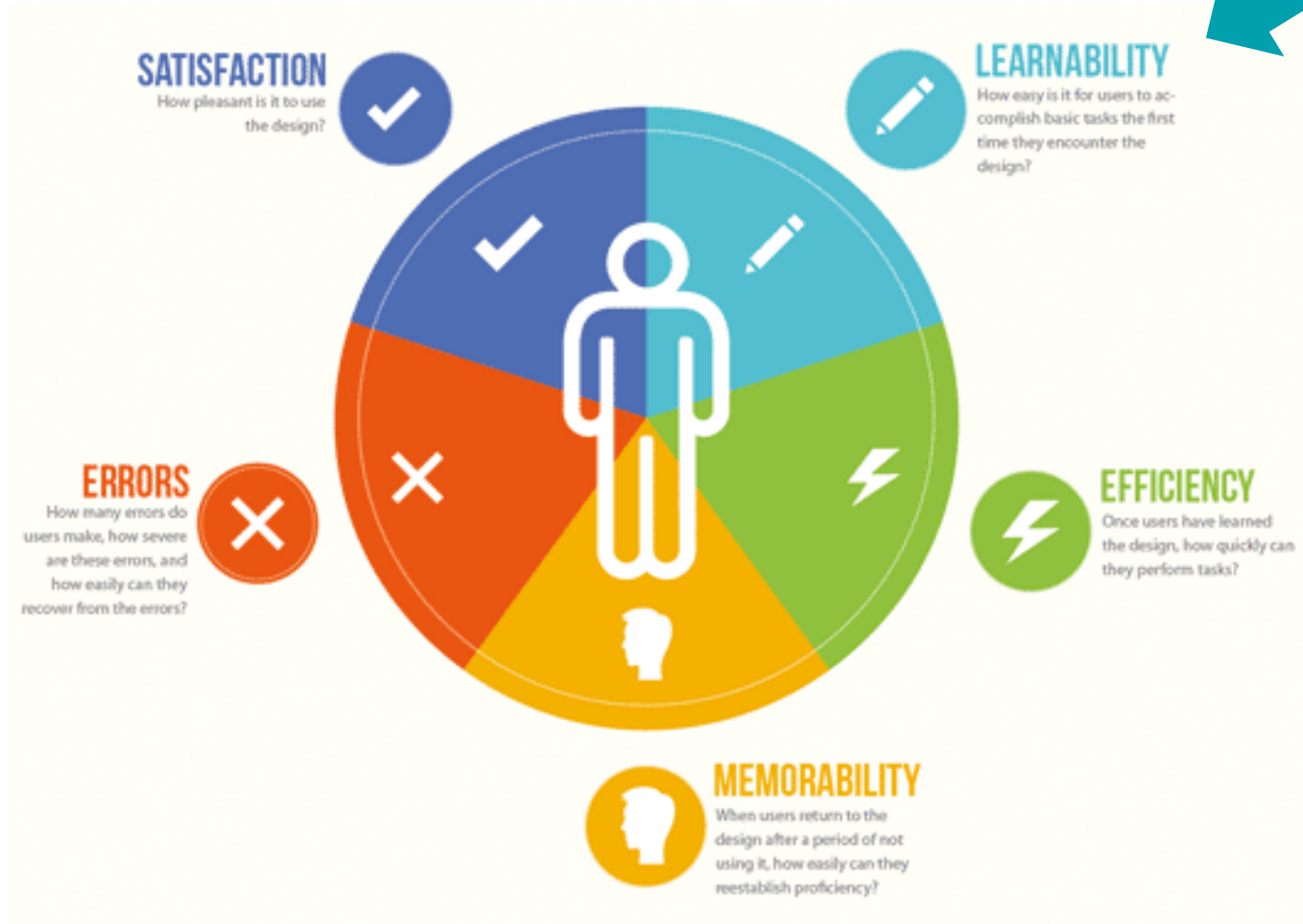
UX IS MORE THAN JUST A NICE UI



USABILITY (PART OF UX)



USABILITY CHARACTERISTICS

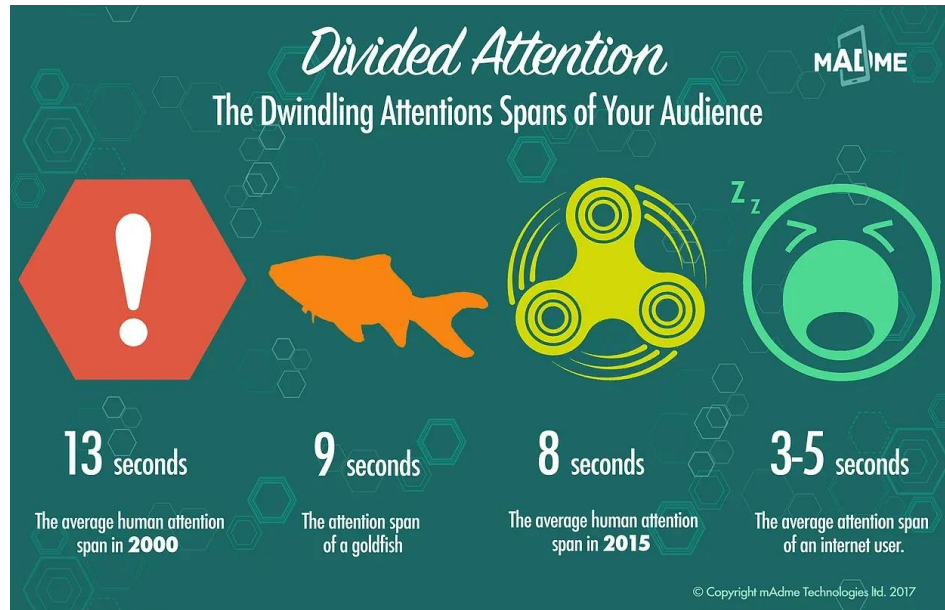




- How easy is it for a user to perform basic tasks or find information when they first open the app?
 - Predictable
 - Feedback: what is the application doing?
 - Use familiar items and icons
 - Consistent and clear layout
 - Limit functionality to the essential only
 - Keep it simple



- Attention span
 - research Microsoft in 2015:



- The user must be able to find the desired information quickly enough or he will drop out



- Place the different UI elements where the user expects them:
 - navigation at the top or on the left
 - navigation preferably not hidden (see trends)
 - search bar at the top and clearly present
 - tooltips

main navigation

utilities

search field

WOMENS MENS

EN | EUR Wish List (0) Sign In My Bag (0)

Search... Q

FORWARD

NEW DESIGNERS CLOTHING SHOES BAGS ACCESSORIES SALE SHOPS

site name

sub navigation

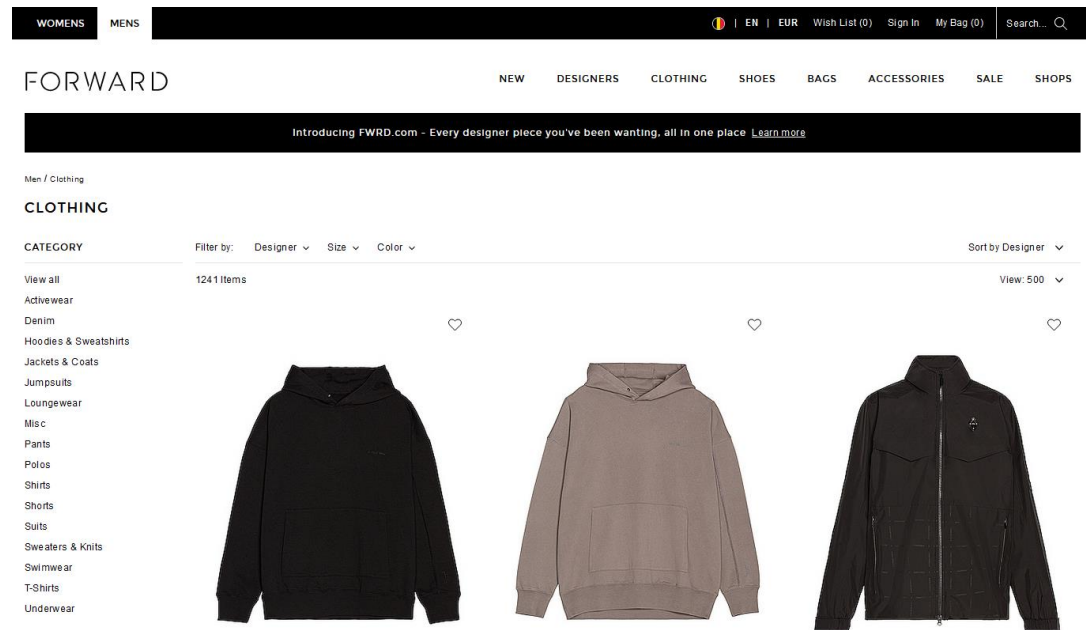
bread crumbs

page title

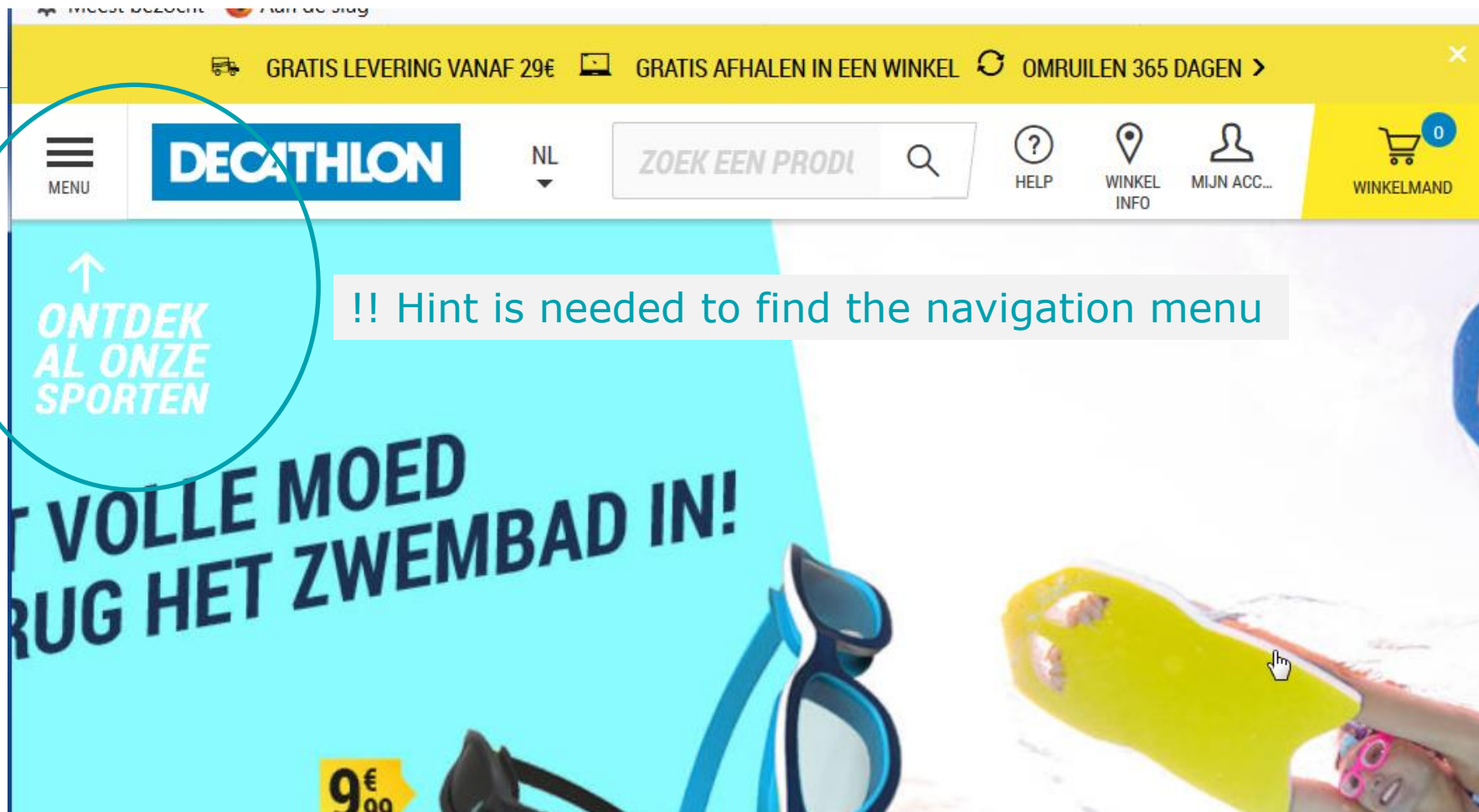
Men / Clothing
CLOTHING

local navigation

- CATEGORY
- View all
 - Activewear
 - Denim
 - Hoodies & Sweatshirts
 - Jackets & Coats
 - Jumpsuits
 - Loungewear
 - Misc
 - Pants
 - Polos
 - Shirts
 - Shorts
 - Suits
 - Sweaters & Knits
 - Swimwear
 - T-Shirts
 - Underwear



www.fwrdd.com



LEARNABILITY: PREDICTABLE



- Make sure it is clear where you can click

< OBVIOUSLY CLICKABLE		REQUIRES THOUGHT >
Click 	Hmm. <i>[Milliseconds of thought]</i> I guess that's a button. Click 	Hmm. Is that a button? 

Don't Make Me Think – Steve Krug

LEARNABILITY: PREDICTABLE



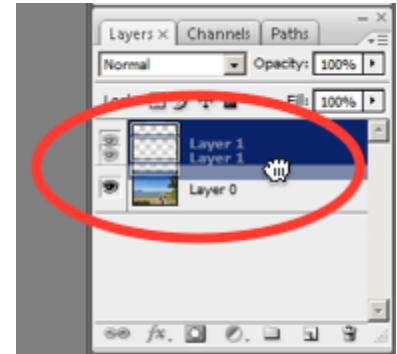
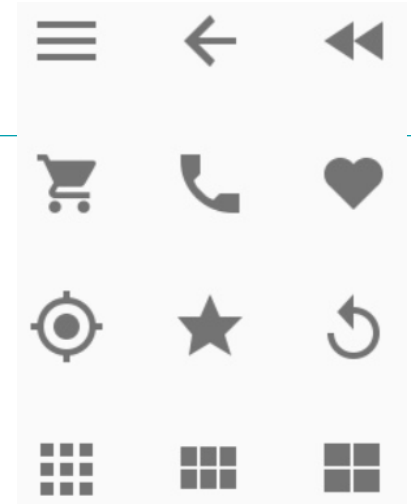
- clear language: watch out for company-specific language or overly technical language



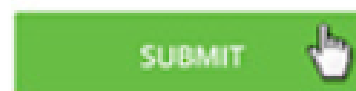
Don't Make Me Think – Steve Krug

LEARNABILITY

- Universal icons:
 - Use icons that are already known
- Universal colour codes:
 - Use red for error messages, ...
- Feedback:
 - Indicate what the application is doing
 - Let the user know if an operation was successful or not



Normal state



Hover state



Active state

LEARNABILITY: LAYOUT



- We don't read, we scan

WHAT WE DESIGN FOR... THE REALITY...

WHAT WE DESIGN FOR...

Read
Read
Read
Read
[Pause for reflection]
Finally, click on a carefully chosen link

THE REALITY...

Look around feverishly for anything that

a) is interesting, or vaguely resembles what you're looking for, and

b) is clickable.

As soon as you find a halfway-decent match, click.

If it doesn't pan out, click the Back button and try again.

On the average web page, users will read at most

28%

of the words during an average visit; 20% is more likely.



The average page visit lasts less than a minute and users often leave web pages in just

10-20 SECONDS



LEARNABILITY: LAYOUT



- Visual hierarchy:
 - the more important, the more visible
 - visualise relations between elements
- Divide page into clear zones

You read this first

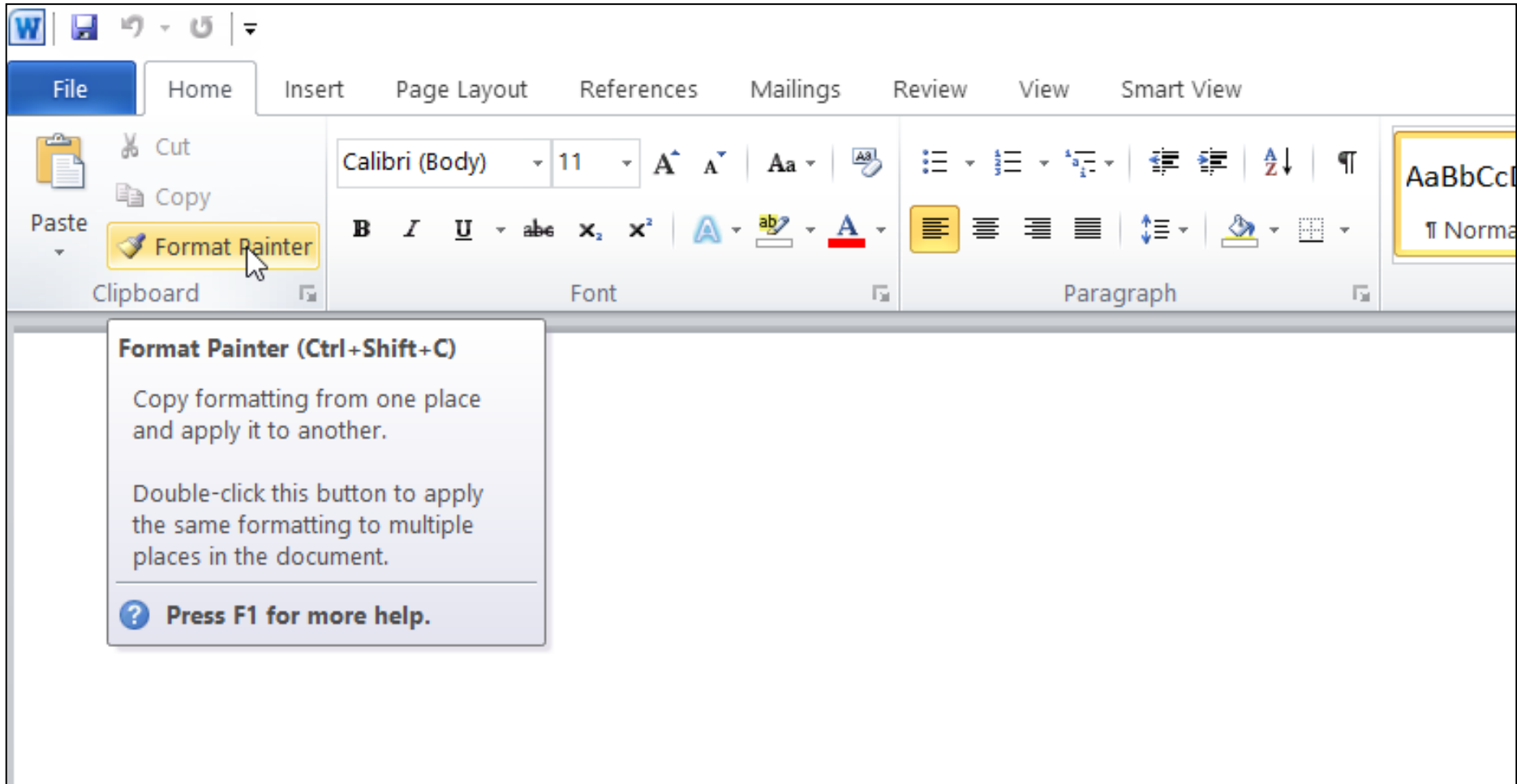
And then you read this.
Maybe someone will read this, sometimes **only the bold part.**

This is Visual Hierarchy, fascinating.

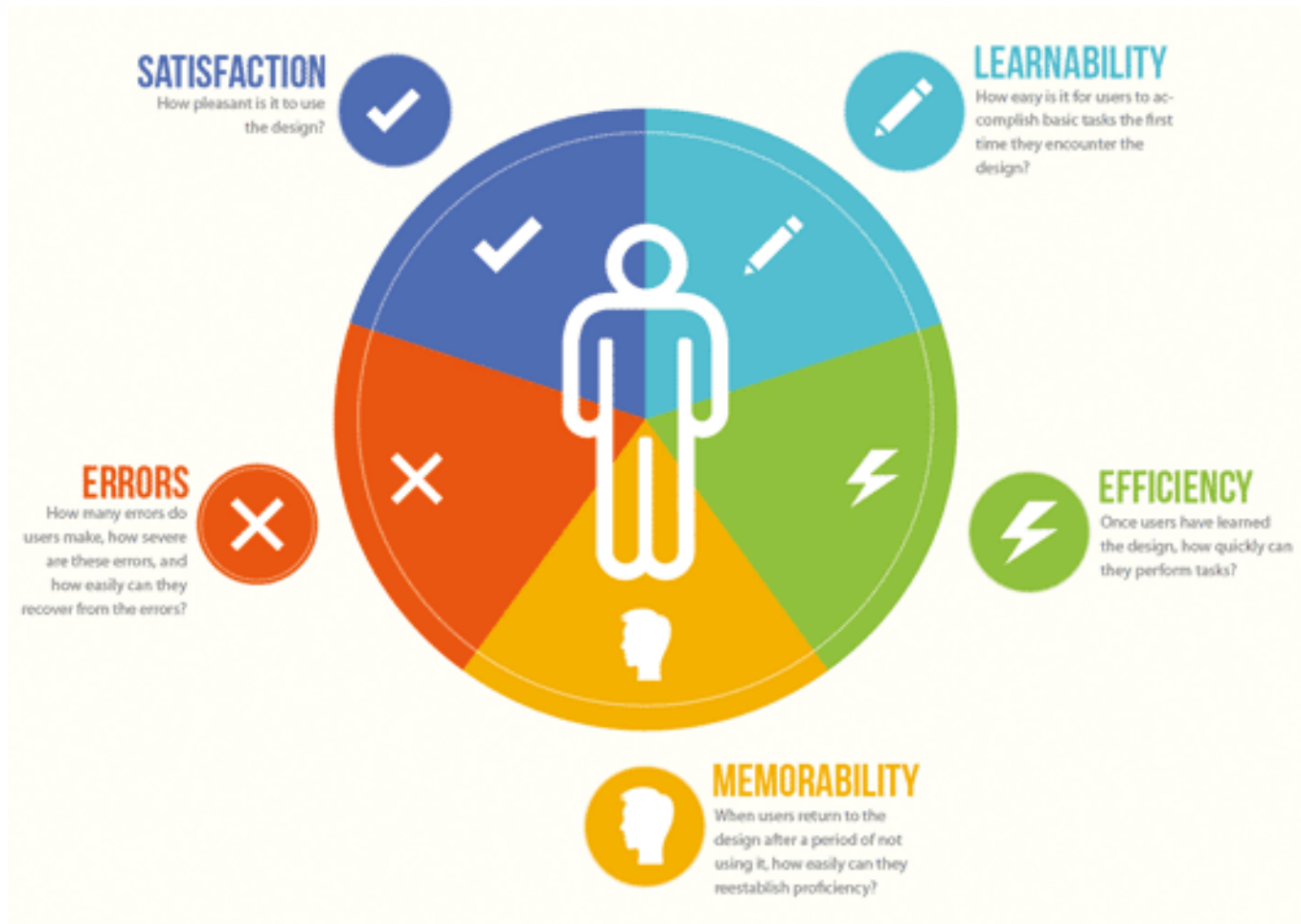
differentiation and hierarchy

no hierarchy

LEARNABILITY: TOOLTIPS



USABILITY CHARACTERISTICS





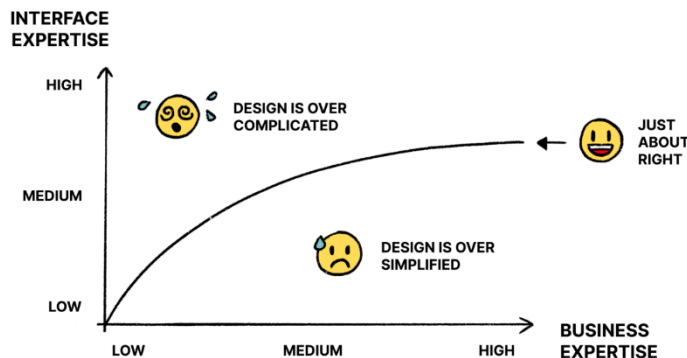
= How fast can a task be performed?
= # of steps needed to perform a task?

- what functionality is important?
- only show what is necessary to perform the desired task quickly
- use default values in fields that usually have the same value





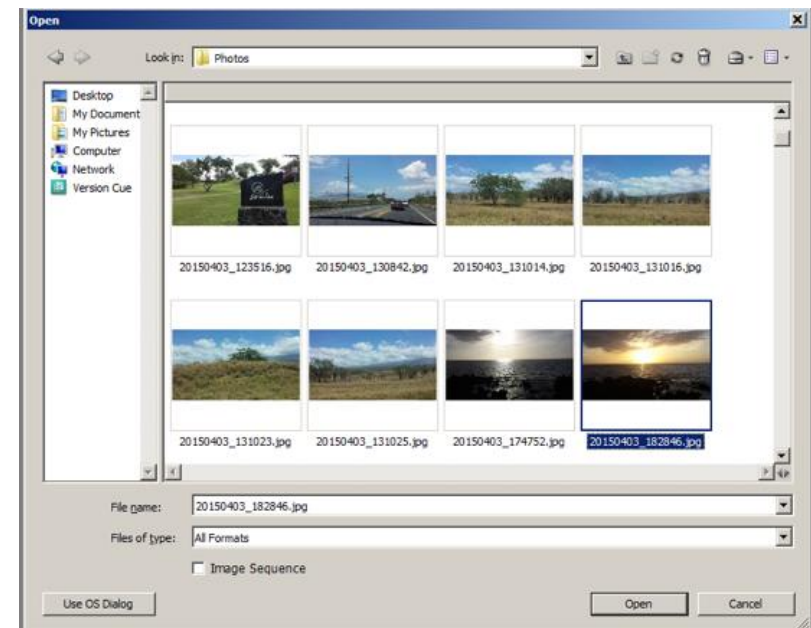
- organize tasks/functions corresponding to tasks on workflow
- provide shortcuts for more experienced users
- adapt the interface complexity to the type of user!
- be careful not to over simplify



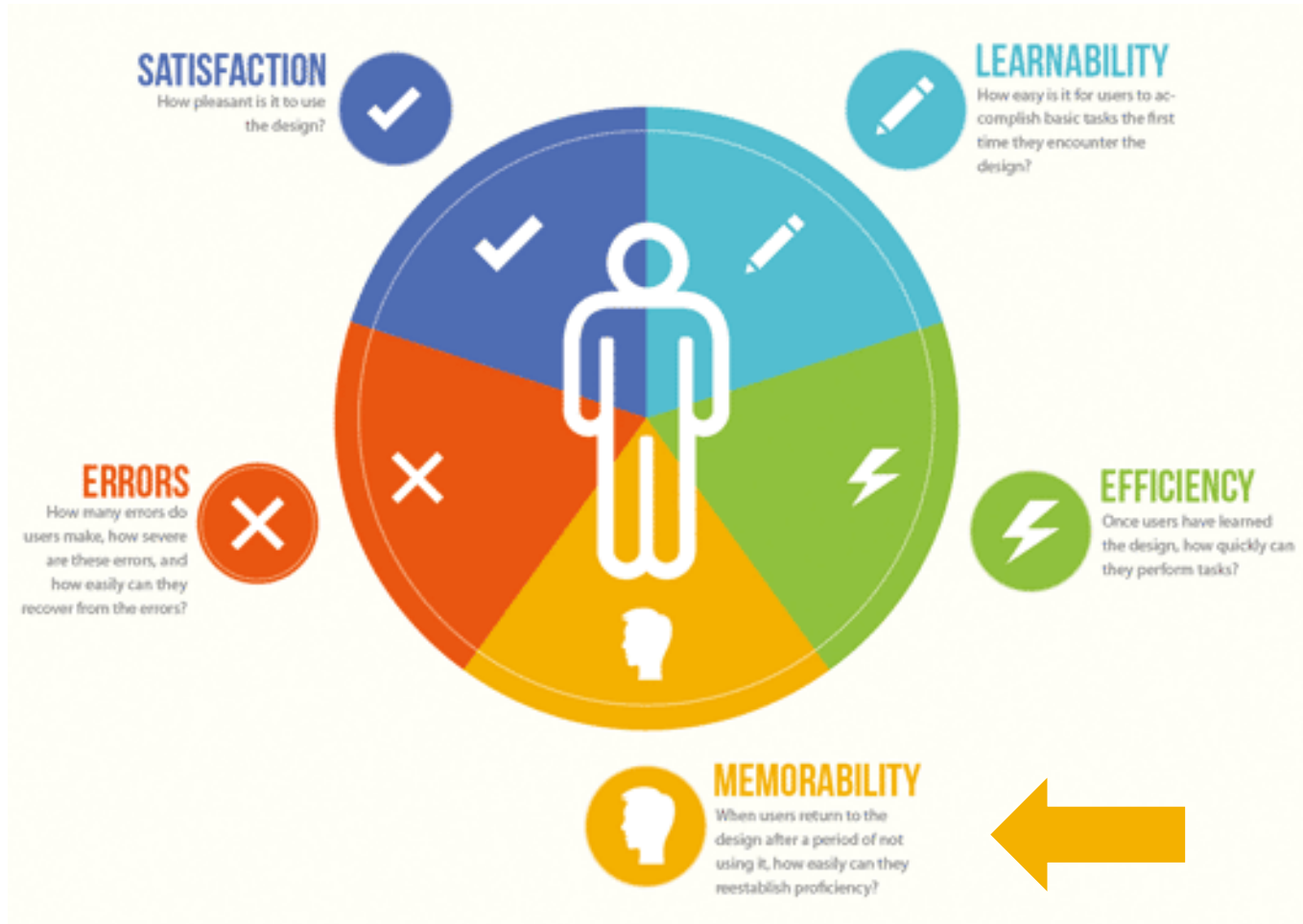


Recognition rather than recall

- Up to 7 items in working memory of human brain
- Keep relevant info on the screen
- Show preview instead of just file name



USABILITY CHARACTERISTICS





When users return to the app after a period of inactivity, how quickly can they get used to the UI again?

≈ learnability and efficiency:

- use of colours e.g. for error messages (red - orange - green)
- use of wizards to perform difficult tasks
- use of tabs

+ but also:

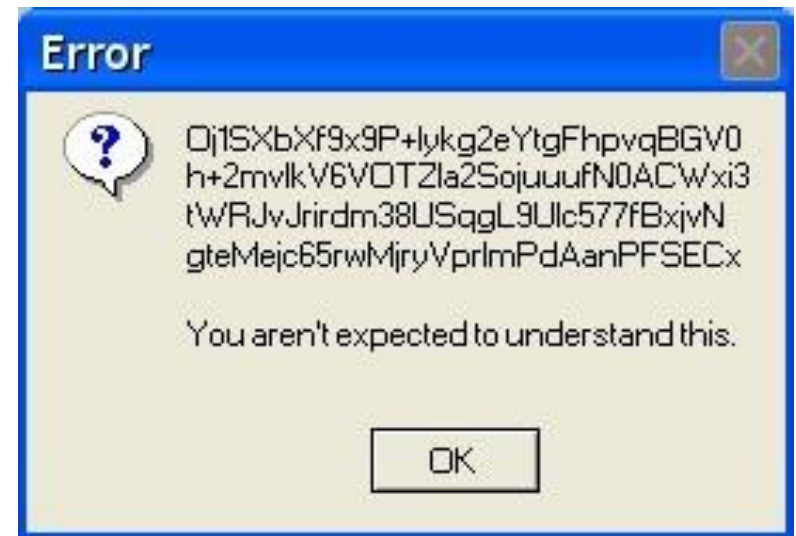
- effects that you don't expect can be easy to remember, especially if they were pleasant

USABILITY CHARACTERISTICS





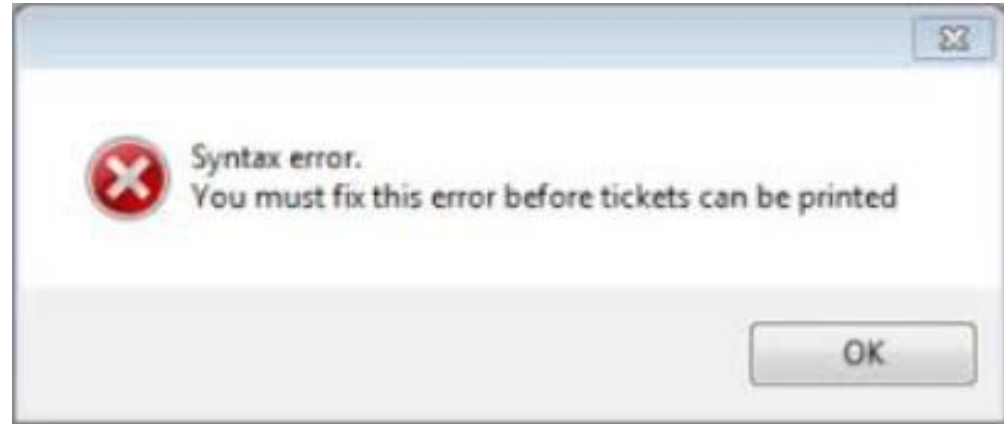
- Helpful error messages:
 - present:
ensure that there is a message every time an error occurs
 - understandable messages





- Helpful error messages:

- specific

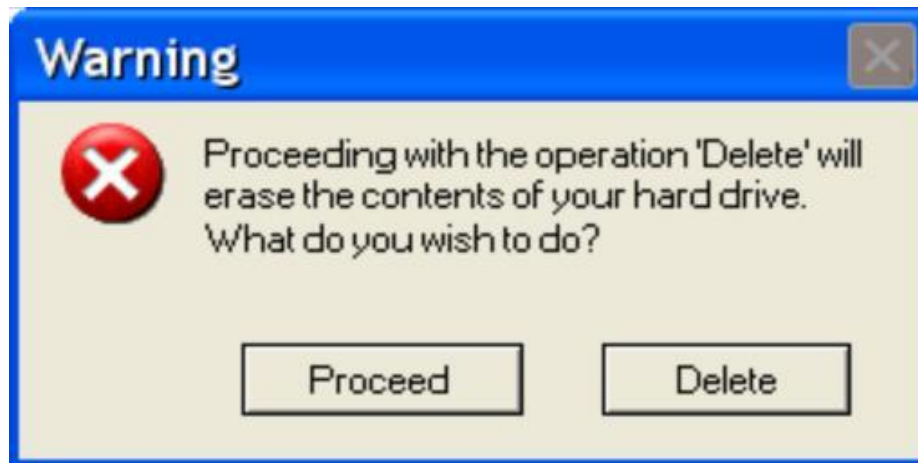


- correct
- consistent:
 - fixed icon
 - fixed layout
 - Always the same display method (popup, fixed position on the screen, ...)

ERRORS



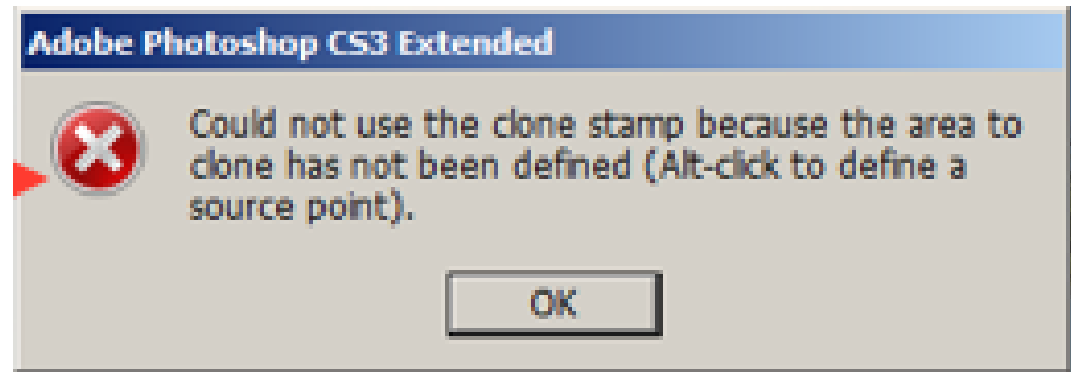
- Helpful error messages:
 - provide help (e.g. how to solve the error?)
 - clear (do not create confusion)



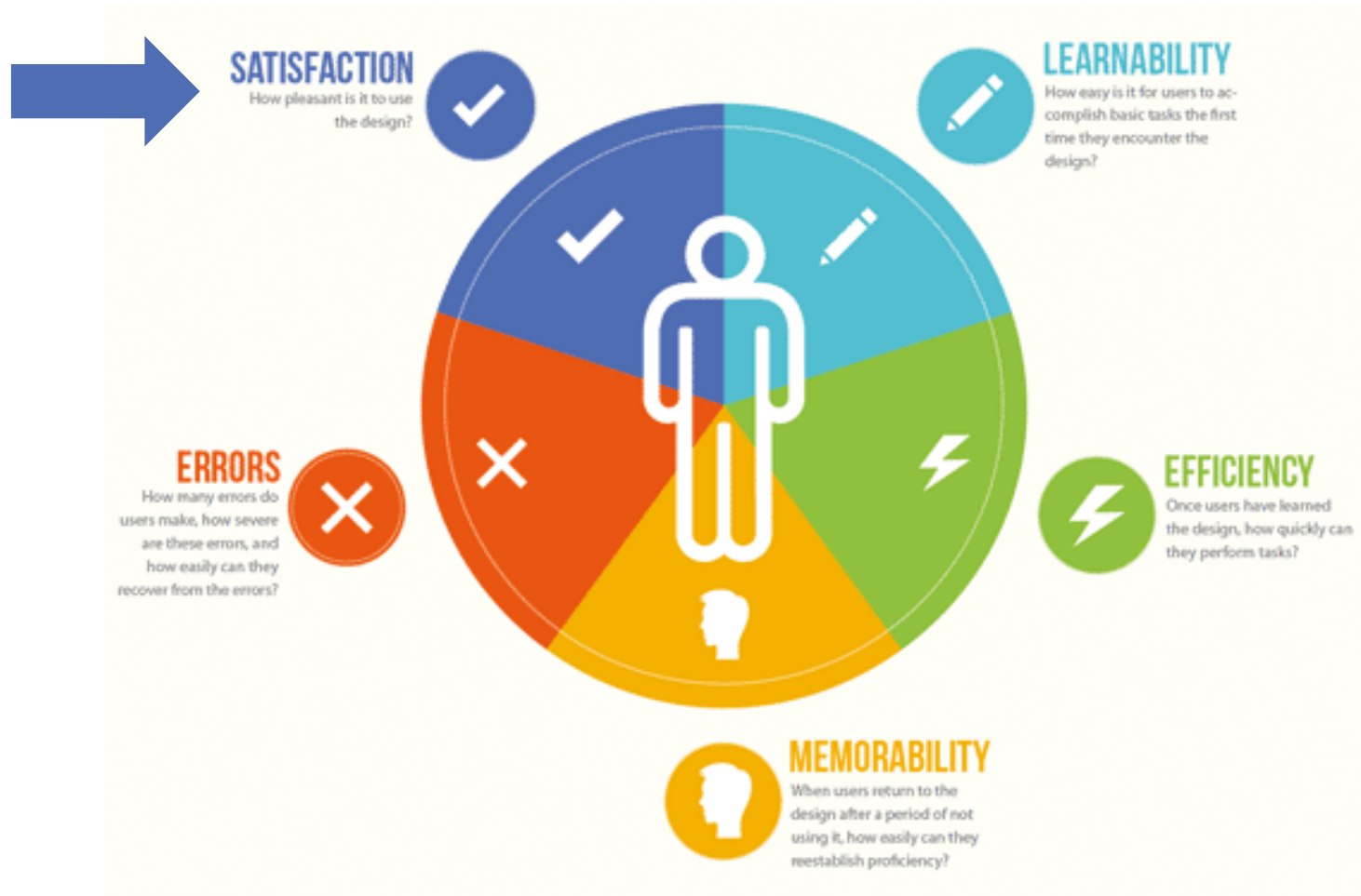
ERROR PREVENTION



- design the system so that errors do not occur frequently
- disable or clearly indicate actions that could cause errors
- extra tooltip for the different selection buttons



USABILITY CHARACTERISTICS



SATISFACTION

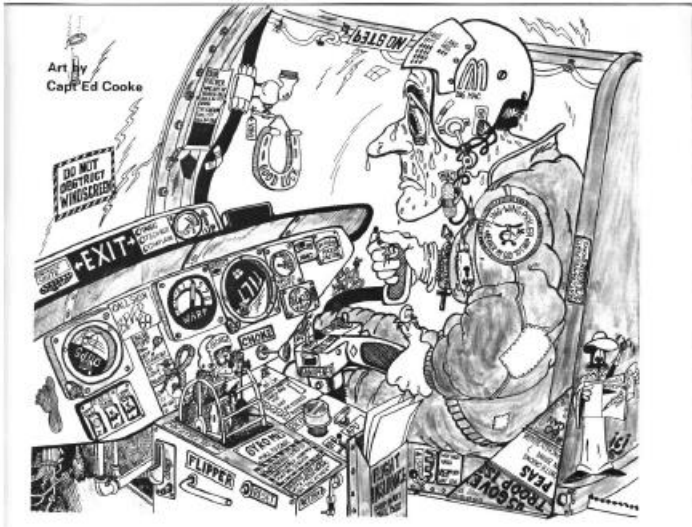


- How enjoyable is it to work with the UI?
 - If a system is easy to learn, has the right features that the user can easily remember, and gives good error messages, a user will be satisfied.
- Looks: does the design look attractive?
- Feel: are the response times short enough?



SAFETY

- User safety:



The GPS said 'turn right'

11 januari 2018 | 02u32



© Foto Toon Royackers - De vrachtwagen nam zijn bocht te kort en kwam met zijn achterwielen eerst in de berm en vervolgens in de gracht terecht.

SAFETY

- Data safety:
 - privacy
 - security
 - hacking



ACCESSIBILITY => WCAG

- Users with special needs:
 - Visual impairments:
 - color blindness: contrast
 - low vision, blindness: screen readers (alt img)
 - hearing impairments:
 - optional subtitles, caption for movies
 - reduced motor skills:
 - Configurable gestures (swiping, tapping, ...)
 - Configurable shortcuts
 - Allow keyboard navigation
 - Learning disorders:
 - font, contrast

ACCESSIBILITY

- Tools:
 - Screen reader for Chrome: [Chromevox](#)
 - Chrome plug-in for visual impairment simulation: [ChromeLens](#)
 - Web Accessibility page analyzer: [Lighthouse](#)
 - App Accessibility page analyzer: [Accessibility Scanner](#)